ChessForensics: Bayesian Detection of Engine Assistance in Competitive Chess

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November 2026

Abstract

I present a statistical framework for detecting engine assistance in competitive chess. The core innovation is a *per-move log-likelihood ratio*: for each position, a logistic regression model fitted on the training corpus predicts the probability that an honest player matches the engine's top choice given position-specific features (evaluation gap, legal move count, complexity, and spread). This position-specific probability feeds a binary log-LR that is summed across moves and integrated over four assistance rates $r \in \{10\%, 25\%, 50\%, 100\%\}$ via a mixture model. Multi-game evidence uses a per-player mixture that charges the model-selection penalty once. Per-band null calibration ensures uniform false-positive rates across rating levels.

Training and evaluation data are strictly separated. Thresholds are selected on 28,670 out-of-sample Chess.com games (V1, 15,586 players) and tested on a held-out set of 22,932 games (V2, 7,505 players, zero overlap). Using band-stratified thresholds, V2's false-positive rate is at or below the target at every level tested (e.g. 0.47% at a 1% target in the 2000–2200 band; 95% CI $[0.38, 0.58]\%$). All ten FIDE-sanctioned cheating cases tested produce positive aggregate evidence (+4.7 to +55.6 dB); eight exceed +10 dB.

A pre-registered head-to-head comparison against four baselines (including a Regan-style two-signal approach) on 5,000 held-out games with paired McNemar tests shows the per-move model achieves 87% per-player detection at 1% FPR for 25%-rate assistance (vs 64% for the Regan-style baseline, $p < 10^{-42}$), while correctly clearing Kasparov's "immortal game" (−4.9 dB) that the Regan-style baseline would flag (+8.4 dB).

Keywords: chess, cheating detection, Bayesian inference, log-likelihood ratio, choice model, engine assistance, fair play.

1 Introduction

The availability of superhuman chess engines on commodity hardware poses a serious and growing threat to competitive chess integrity. A player who secretly consults an engine—even on a small fraction of moves—gains a decisive advantage that is invisible to opponents and arbiters. High-profile cases, from Borislav Ivanov's implausible results at the 2012 Zadar Open to the Carlsen–Niemann controversy at the 2022 Sinquefield Cup, have underscored the need for rigorous statistical methods.

The foundational work is due to Regan [Regan and Haworth, 2011, Regan, 2014], who introduced the multinomial choice model that maps engine evaluations to expected move probabilities, then tests whether a player's accuracy is statistically anomalous. The present system builds on Regan's framework with several extensions:

1. **Per-move log-likelihood ratios.** Rather than aggregating moves into a single game-level z-score, the system computes a binary log-LR for each move using a position-specific probability of matching the engine's top choice. This probability comes from a per-band

logistic regression on six position features (evaluation gap, legal move count, complexity, spread, gap-small indicator, only-move indicator), fitted on 2.7 million training games (AUC ≈ 0.75).

2. **Four-rate mixture model** over assistance levels $r \in \{10\%, 25\%, 50\%, 100\%\}$. Per-move log-LRs are computed separately for each rate and combined via logsumexp, enabling detection of partial or intermittent engine use without requiring the rate to be specified in advance.
3. **Per-band null calibration** that z-normalises the raw dB score against each rating band's honest distribution, ensuring uniform false-positive rates across rating levels.
4. **Per-player aggregation** that sums per-rate log-LRs across games and applies the mixture penalty once, dramatically improving multi-game detection of partial cheating.
5. **Fifteen supplementary signals** spanning move accuracy, error distribution shape, clock behaviour, and engine fingerprinting, available for display and secondary analysis.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 presents the choice model and signal definitions. Section 4 describes the end-to-end analysis pipeline. Section 5 describes null calibration. Section 6 details the Bayesian evidence framework. Section 7 describes corpus construction and train/test separation. Section 8 reports validation results. Section 9 presents a pre-registered head-to-head comparison against four baselines. Section 10 discusses limitations, and Section 11 concludes.

2 Related Work

Regan and Haworth [Regan and Haworth, 2011] introduced the *intrinsic performance rating* (IPR), estimating a player's strength from the sensitivity parameter of a multinomial logit model fitted to engine evaluations. Regan [Regan, 2014] subsequently developed a hypothesis-testing framework comparing observed match rates against expected rates under a Poisson-binomial model, applying this methodology as a consultant for FIDE's Anti-Cheating Commission.

Other approaches include naïve centipawn-loss thresholds [Guid and Bratko, 2006], which suffer from position-complexity confounding and lack formal error control; large-scale historical analyses using aggregate statistics [Alliot, 2017, Ferreira, 2013]; move-matching percentages without per-position calibration; and machine-learning classifiers trained on labelled data, which risk overfitting to specific engine versions.

This work differs in two key respects. First, instead of testing aggregate game-level statistics, I compute a *per-move* log-likelihood ratio using position-specific match probabilities from a fitted logistic model, which preserves position-level information that game-level z-scores discard. Second, the per-move log-LRs are integrated over a four-rate mixture model for partial assistance, with per-band null calibration from a 3.4-million-game corpus under strict train/test separation. I validate against ten FIDE-sanctioned cheating cases with known ground truth, 28,670 out-of-sample Chess.com games, and a pre-registered head-to-head comparison against Regan-style baselines with paired statistical tests.

3 The Choice Model and Signal Extraction

3.1 Multinomial Choice Model

Given a position with K legal moves ranked by engine evaluation, let $\delta_i \geq 0$ denote the evaluation loss (in pawns) of move i relative to the best move ($\delta_1 = 0$). The probability that a player of sensitivity $s > 0$ and consistency $c > 0$ plays move i is

$$P(i \mid s, c) = \frac{\exp\bigl(-(\delta_i/s)^c\bigr)}{\sum_{j=1}^{K} \exp\bigl(-(\delta_j/s)^c\bigr)}, \tag{1}$$

a softmax (multinomial logit) model [Regan and Haworth, 2011] where smaller s corresponds to stronger play (finer sensitivity to evaluation differences) and larger c implies sharper discrimination between small and large errors.

Tail extrapolation. Engine analysis returns the top k moves (MultiPV = 5). For the remaining $K - k$ legal moves, I extrapolate evaluation losses linearly: $\delta_{k+j} = \delta_k + j \cdot \Delta$, where $\Delta = \max(0.05, \delta_k - \delta_{k-1})$, clipped at $\delta_{\max} = 6.0$ pawns. This prevents upward bias in the fitted s by maintaining the proper normaliser.

Maximum-likelihood estimation. Parameters (s, c) are fitted by maximising the log-likelihood

$$\ell(s, c) = \sum_{m=1}^{n} \log P\big(\text{played}_m \mid s, c\big) \tag{2}$$

over n scoreable positions, using L-BFGS-B [Byrd et al., 1995] in log-space $(\log s, \log c)$ with multiple restarts. Standard errors are obtained from the observed Fisher information (numerical Hessian) and back-transformed via the delta method: $\text{SE}_s = s\sqrt{\widehat{\text{Var}}[\log s]}$.

Intrinsic Performance Rating (IPR). A calibration curve $\{(\bar{E}_b, \tilde{s}_b)\}_{b=1}^{B}$ maps each rating band's median fitted s to its midpoint Elo $\bar{E}_b$. Monotonicity is enforced via isotonic regression (pool-adjacent-violators algorithm). For a subject game with fitted $\hat{s}$, the IPR is obtained by log-linear interpolation:

$$\widehat{\text{Elo}} = \text{interp}\big(\log \hat{s};\ \{\log \tilde{s}_b\},\ \{\bar{E}_b\}\big). \tag{3}$$

Confidence intervals propagate SE_s through the same interpolation. The curve is validated on the 12 band midpoints (Section 8).

3.2 Match-Rate Signals

For each scoreable position m, the reference model yields per-position match probabilities $p_m^{(1)}, p_m^{(2)}, p_m^{(3)}$ for agreeing with the engine's top-1, top-2, and top-3 moves respectively. The observed match count $X = \sum_m \mathbf{1}[\text{matched}]$ follows a *Poisson-binomial* distribution [Hong, 2013] with heterogeneous success probabilities, for which exact computation is available via $O(n^2)$ dynamic programming.

Definition 1 (Match-rate z-score)**.**

$$z_{tk} = \frac{X_k - \sum_m p_m^{(k)}}{\sqrt{\sum_m p_m^{(k)}(1 - p_m^{(k)})}}, \qquad k \in \{1, 2, 3\}. \tag{4}$$

The exact one-sided p-value $\Pr(X_k \geq x_k)$ is computed from the Poisson-binomial PMF; the mid-p variant $\Pr(X_k > x_k) + \frac{1}{2}\Pr(X_k = x_k)$ is used to reduce conservatism.

3.3 Accuracy Signals

Average centipawn loss (ACPL). Let δ_m^{cp} denote the centipawn loss on move m. Under the reference model, the per-position expected loss $\mu_m = \mathbb{E}[\delta_m^{\text{cp}}]$ and variance $\sigma_m^2 = \text{Var}[\delta_m^{\text{cp}}]$ are computed from the multinomial choice probabilities. The ACPL signal tests whether the observed mean loss is anomalously low:

$$z_{\text{acpl}} = \frac{\bar{\delta}^{\text{cp}} - \bar{\mu}}{\sqrt{\sum_m \sigma_m^2 / n^2}}, \tag{5}$$

where lower (more negative) z indicates suspiciously accurate play.

Accuracy–complexity slope. A logistic regression of matched_t1 on position complexity yields a slope; stronger players (and engines) maintain accuracy even on complex positions. The slope z-score tests whether the observed slope is anomalously flat relative to the band reference.

CPL–complexity slope. Analogously, a linear regression of centipawn loss on complexity yields a slope; engine-assisted play shows unnaturally constant CPL across complexity levels.

CPL tail test. Rather than testing the mean, this signal tests whether the blunder distribution is suspiciously *truncated*—i.e., whether large errors are missing. Per position, the probability of a blunder exceeding threshold β (default 200 cp) is

$$P(\text{blunder}_m) = 1 - \Phi\left(\frac{\beta - \mu_m}{\sigma_m}\right). \tag{6}$$

The observed blunder rate is compared to the expected rate via a variance-weighted z-test, with sign inverted so that fewer blunders yields positive z.

3.4 Timing Signals

Clock data, when available, provides signals orthogonal to move quality. Each timing signal carries a validity weight $w_{\text{speed}} \in [0, 1]$ that depends on time control: classical receives full weight ($w = 1.0$), rapid $w = 0.75$, blitz $w = 0.30$, bullet $w = 0.05$, and correspondence $w = 0$.

Time–complexity correlation. Honest players spend more time on complex positions. The Spearman rank correlation ρ between think times and position complexity is computed; $z_{\text{rho}} = -\rho/\text{SE}_\rho$, so that low correlation (engine-like uniform speed) yields positive z.

Residual coefficient of variation. After regressing log(time) on complexity, the standard deviation of residuals measures timing uniformity. Lower residual CV (more robotic timing) is suspicious.

Cadence detection. A modal-clustering analysis identifies the fraction of moves within ± 0.35 s of the most common think time. High cadence fraction (e.g., consistently playing on the increment) is suspicious.

Timing–accuracy cross-signal. Among high-complexity positions (top 40%) where the player matched the engine's top move, the mean think time is compared to the band baseline. Suspiciously fast correct play on hard positions is flagged via a log-space z-test.

3.5 Structural Signals

Switchpoint detection. A two-rate binomial likelihood-ratio scan tests whether a player's accuracy undergoes a sudden jump mid-game, consistent with starting engine consultation after an initial honest phase. For each candidate split point k (with minimum segment length 8), I compute

$$\Lambda_k = \ell(\hat{p}_{\text{before}}, \hat{p}_{\text{after}}) - \ell(\hat{p}_{\text{all}}), \tag{7}$$

where $\hat{p}$ are the MLE binomial rates. The best split $k^* = \arg\max_k \Lambda_k$ is tested via 2,000 permutations, requiring both $p < 0.05$ and a rate jump exceeding 18 percentage points. The z-score is the probit transform of the permutation p-value, boosted by a factor of 1.3 when the player was losing at the switchpoint.

IPR gap. The difference between the intrinsic performance rating and the player's nominal Elo, normalised by the estimated single-game IPR standard deviation (≈ 250 Elo), provides a signal that is strongest for lower-rated players receiving full engine assistance.

Engine fingerprinting. When a second engine (e.g., Leela Chess Zero at low depth) is available, the *deep-only match rate*—agreement with the primary engine on positions where the second engine disagrees—isolates engine-specific move selection patterns that honest players are unlikely to replicate.

4 Analysis Pipeline

A game is processed through the following stages:

1. Engine analysis (three passes). Every position is first analysed at depth 4/MultiPV 3 ("L0"), which identifies moves that any shallow search would find. A depth-12/MultiPV-5 pass then scores all positions. A priority ranker flags critical positions—where the played move diverges from the engine's top choice, where the evaluation swings, or where the position is tactically complex—and promotes them to a depth-22/MultiPV-5 deep pass, up to a budget of $\sim$40% of total plies. This adaptive schedule concentrates compute on the positions that carry forensic information.

2. Position filtering. Positions are excluded from scoring if they fall within detected book lines or are forced (one legal move). Positions with evaluation beyond ± 6 pawns are also excluded—not because cheating is impossible there, but because the choice model's discriminative power vanishes: when any reasonable move wins, the expected top-1 match rate approaches 100% for honest players and cheaters alike, so including these positions adds noise without signal. A minimum of 8 scoreable positions is required; games below this threshold receive no verdict.

3. Feature extraction. For each scoreable position, the system records: whether the played move matches the engine's top-1/2/3 choices, the centipawn loss, the position complexity (number of reasonable alternatives weighted by evaluation spread), clock time consumed, and the L0 agreement flag. These per-move features are assembled into a DataFrame from which game-level signals are computed.

4. Signal computation. Fifteen signals are extracted (Section A): three match-rate z-scores via Poisson-binomial tests, an ACPL z-score, two accuracy-vs-complexity slopes, a CPL tail test, a switchpoint scan, an IPR gap, four timing signals (including a timing–accuracy cross-signal), and two engine-fingerprint signals (when a second engine is available).

5. Per-move scoring and evidence combination. For each scoreable position, a per-band logistic regression predicts the probability of matching the engine's top choice. A binary log-LR is computed per move and summed across positions, integrated over four assistance rates via the mixture model, and calibrated against the band's honest null distribution (Section 6). A Monte Carlo sensitivity band (4,000 draws) quantifies how the posterior moves under model uncertainty.

6. Multi-game aggregation. When multiple games from the same player are available, per-rate log-LRs are summed across games and the mixture penalty is applied once (Section 6.6). The aggregate posterior and per-game evidence breakdown constitute the final verdict.

5 Null Calibration

Raw z-scores from the hypothesis tests of Section 3 exhibit systematic bias and overdispersion relative to the standard normal, due to model misspecification and the simplifying assumptions of the choice model. These are corrected via a three-tier calibration scheme.

Tier 1: Per-band calibration. For each of 12 rating bands (width 200 Elo, from 800–1000 to 2800+), the empirical mean (bias b_{band}) and standard deviation (scale σ_{band}) of each signal's z-scores on the corpus of honest games within that band. The corrected z-score is

$$z_{\text{cal}} = \frac{z_{\text{raw}} - b_{\text{band}}}{\sigma_{\text{band}}}. \tag{8}$$

Tier 2: Pooled calibration. When the target band has insufficient data for a particular signal, the system falls back to the pooled bias and scale computed across all bands (e.g., ACPL: $b = 0.27$, $\sigma = 1.99$ from 70,948 player-games).

Tier 3: Empirical defaults. When no corpus calibration is available (e.g., a baseline built without clock data), hardcoded empirical values are used. These were measured once on a large sample and serve as a fallback (e.g., time_uniformity: $b = -0.09$, $\sigma = 1.40$). The Tier 3 values for accuracy signals (e.g., t1_match: $b = -0.51$, $\sigma = 1.66$) are more conservative than the per-band Tier 1 values (typically $b \approx 0$, $\sigma \approx 1.1$–1.2) because they pool across all ratings, absorbing inter-band variance into a wider scale.

This three-tier design ensures graceful degradation: the system produces calibrated posteriors even with incomplete baselines, while exploiting the most specific calibration data when available.

6 Bayesian Evidence Combination

6.1 Per-Move Log-Likelihood Ratio

The central innovation of this system is a *per-move* scoring mechanism. Rather than summarising a game into aggregate statistics (match rate, ACPL) and testing those, I compute a log-likelihood ratio for each individual move, preserving position-level information.

For each scoreable position m, let $t_m \in \{0, 1\}$ indicate whether the player matched the engine's top choice, and let p_m denote the position-specific probability that an honest player of the appropriate rating would make this match. Under the null (honest play), the probability of the observed outcome is

$$P(t_m \mid H_0) = p_m^{t_m}(1 - p_m)^{1-t_m}. \tag{9}$$

Under the alternative at assistance rate r (the player consults the engine on each move independently with probability r):

$$P(t_m \mid H_1, r) = \begin{cases} r + (1 - r)\, p_m & \text{if } t_m = 1, \\ (1 - r)(1 - p_m) & \text{if } t_m = 0. \end{cases} \tag{10}$$

The per-move log-LR at rate r is therefore:

$$\ell_m(r) = \begin{cases} \log\big(r + (1 - r)\, p_m\big) - \log p_m & \text{if } t_m = 1, \\ \log(1 - r) & \text{if } t_m = 0. \end{cases} \tag{11}$$

Position-specific p_m via logistic regression. The key to this approach is an accurate, well-calibrated estimate of p_m. Rather than using the global Poisson-binomial model (Section 3.2), which assigns expected match rates based on the choice model fitted to band-level data, I train a per-band logistic regression [Hosmer et al., 2013] on six position features:

$$\operatorname{logit}(p_m) = \beta_0 + \beta_1 \log(1 + g_m/100) + \beta_2 \log L_m + \beta_3 C_m + \beta_4 S_m + \beta_5 G_m + \beta_6 O_m, \tag{12}$$

where g_m is the evaluation gap (centipawns) between the top two engine moves, L_m the number of legal moves, C_m the position complexity, S_m the evaluation spread among top candidates, G_m the fraction of candidates within 30 cp of the best, and O_m an indicator for forced moves ($L_m \leq 2$ or gap > 3 pawns). A separate logistic regression is fitted for each of 12 rating bands on the training corpus (2.7 million games), achieving AUC ≈ 0.75 with well-calibrated probability outputs.

6.2 Mixture Model Over Assistance Rates

A cheater may not use the engine on every move. Following the finite-mixture framework [McLachlan and Peel, 2000], I model four assistance rates $r \in \mathcal{R} = \{0.10, 0.25, 0.50, 1.0\}$ with a uniform prior $\pi_r = 1/|\mathcal{R}|$. The per-game log-LR at rate r sums over all n scoreable positions:

$$\log \mathrm{LR}_{\mathrm{game}}(r) = \sum_{m=1}^{n} \ell_m(r). \tag{13}$$

The marginal log-LR integrates over rates:

$$\log \mathrm{LR}_{\mathrm{mix}} = \log \sum_{r \in \mathcal{R}} \pi_r \cdot \mathrm{LR}_{\mathrm{game}}(r) = \operatorname{logsumexp}\big(\{\log \mathrm{LR}_{\mathrm{game}}(r) + \log \pi_r\}_{r \in \mathcal{R}}\big). \tag{14}$$

6.3 Per-Band Null Calibration

The raw per-move dB score exhibits systematic band-dependent bias: stronger players have higher baseline match rates, and the logistic model, while well-calibrated within each band, does not perfectly cancel rating effects across bands. I correct this by z-normalising against each band's honest distribution:

$$\mathrm{dB}_{\mathrm{cal}} = \frac{\mathrm{dB}_{\mathrm{raw}} - b_{\mathrm{band}}}{\sigma_{\mathrm{band}}} \cdot \sigma_{\mathrm{pool}} + b_{\mathrm{pool}}, \tag{15}$$

where $(b_{\mathrm{band}}, \sigma_{\mathrm{band}})$ are the mean and standard deviation of honest dB scores in the target band, and $(b_{\mathrm{pool}}, \sigma_{\mathrm{pool}})$ are the pooled reference. This ensures that a single global threshold achieves uniform FPR across rating bands.

6.4 Posterior Probability

With a base-rate prior $\pi_0 = 0.01$ (1% prevalence of engine assistance), the posterior is

$$P(\text{cheating} \mid \text{game}) = \operatorname{sigmoid}\big(\operatorname{logit}(\pi_0) + \log \mathrm{LR}_{\mathrm{mix}}\big), \tag{16}$$

where $\operatorname{logit}(p) = \log \frac{p}{1-p}$ and $\operatorname{sigmoid}(x) = \frac{1}{1+e^{-x}}$. Evidence strength is also reported in decibans [Good, 1979]:

$$\mathrm{dB} = \frac{10 \cdot \log \mathrm{LR}}{\ln 10}. \tag{17}$$

6.5 Sensitivity Band

A Monte Carlo procedure (4,000 draws) perturbs the per-move p_m predictions by their estimated uncertainty (5% relative jitter when the corpus exceeds 50 games, 10% otherwise) and recomputes the posterior, yielding a 5th–95th percentile band that reflects model uncertainty.

6.6 Multi-Game Per-Player Aggregation

For N games by the same player, the per-rate log-LRs are summed across games:

$$L_r = \sum_{g=1}^{N} \log \mathrm{LR}_g(r), \qquad \log \mathrm{LR}_{\text{player}} = \mathrm{logsumexp}\big(\{L_r + \log \pi_r\}_r\big). \tag{18}$$

This formulation assumes a *single constant assistance rate* per player across all games, which charges the mixture penalty $\log |\mathcal{R}| \approx 6\,\mathrm{dB}$ exactly once rather than once per game. This is critical for detecting partial assistance: in the Feller case (Section 8.5), per-game analysis misses the signal, while per-player aggregation yields +20.8 dB (detected).

6.7 Supplementary Signal-Level Scoring

In addition to the per-move LR, the system computes fifteen game-level signals (Section A) spanning match rates, ACPL, accuracy slopes, timing, switchpoint detection, IPR gap, and engine fingerprinting. These can be combined via a multivariate Gaussian log-LR with correlation adjustment and NNLS projection (detailed in Section 9 as baseline method M4). The signal-level approach is retained for display, secondary analysis, and comparison purposes, but the per-move LR is the primary scoring mechanism for all operational verdicts.

7 Corpus and Calibration Data

7.1 Corpus Construction

The reference corpus comprises 3,400,451 rated rapid and classical games (6.78 million player-games, 229.6 million positions) drawn from two sources: the Lichess open database [Lichess.org, 2024] supplies the bulk of games in bands below Elo 2200, while TWIC (The Week in Chess) OTB tournament archives [Crowther, 2024] supply the majority of games in the 2200–2400 through 2800+ bands. Collection filters:

- Minimum 20 plies per game.
- Maximum 300 Elo gap between opponents.
- Rated games only; no correspondence.
- Accounts flagged for terms-of-service violations excluded.

A deep engine pass (Stockfish 17 [Stockfish Developers, 2024], depth 22, MultiPV 5) was applied to 41,698 games (70,906 player-games, 3.36 million positions). A "lite" pass incorporating Lichess server evaluations extended ACPL statistics to 226 million positions.

7.2 Rating Bands

Twelve bands span the rating range: a catch-all below 800, ten bands of width 200 Elo from 800–1000 through 2600–2800, and one open-ended band above 2800. Each band stores the median choice-model parameters $(\tilde{s}_b, \tilde{c}_b)$, expected match rates, ACPL distribution, accuracy–complexity slopes, and timing baselines. Per-band null calibration (Section 5) is computed from the games within each band.

7.3 Train/Test Independence

The corpus is partitioned into a training set (used for fitting the choice model, computing per-band calibration, and estimating the correlation matrix) and a held-out test set of 682,952 games used exclusively for false-positive rate estimation and calibration verification. No test-set game contributes to any model parameter. Out-of-sample titled-player games (5,560 games from 92 GM/IM/WGM accounts, Section 8.2) were collected from the Lichess public API after the corpus

Table 1: Signal effect sizes under engine assistance. Measured values are mean z-scores from 11 strong cheater games (posterior $> 5\%$) among 55 cheater player-games from 87 banned Lichess accounts. Signals with fewer than 5 observations retain prior estimates.

Signal	Family	N_{obs}	Measured $\bar{z}$	Adopted μ
t1_match	accuracy	11	+3.30	3.3
t2_match	accuracy	11	+2.23	2.2
t3_match	accuracy	11	+2.09	2.1
acpl	accuracy	11	+1.90	1.9
cpl_tail	shape	7	+2.18	2.2
slope	shape	9	+0.29	0.3
cpl_slope	shape	9	+0.00	0.0
time_uniformity	timing	11	+1.30	1.3
time_complexity	timing	11	+0.71	0.7
time_cadence	timing	11	+0.47	0.5
timing×accuracy	timing	8	−0.10	0.0
ipr_gap	strategic	1	—	2.0*
switchpoint	strategic	2	—	2.0*
deep_only	engine	0	—	2.4*
engine_fp	engine	0	—	2.5*

*Insufficient observations; prior estimate retained.

snapshot and share no games with the calibration partition. All ten cheater case PGNs originate from independent sources (TWIC archives, FIDE tournament bulletins) that share no games with the calibration corpus.

7.4 Corpus Integrity Audit

To quantify contamination by undetected cheaters, I queried the Lichess account-status API for all 11,873 distinct users in the corpus. Of these, 87 (0.73%) were flagged for terms-of-service violations, corresponding to an estimated 491 contaminated games (0.068% of the corpus). This contamination rate is negligible: even if all flagged games exhibited maximum engine assistance, their effect on the per-band calibration statistics (computed over thousands of games per band) would be undetectable.

7.5 Effect Sizes from Banned Accounts

To ground the alternative-hypothesis effect vector $\boldsymbol{\mu}_{1.0}$ in empirical data rather than theoretical assumptions, I analysed games from all 87 flagged accounts. Of 55 cheater player-games, 11 showed strong evidence of engine assistance (posterior $> 5\%$). Table 1 reports the measured effect sizes alongside the values adopted in the model.

The data confirm that **match-rate signals dominate** detection (t1: $+3.3\sigma$, ACPL: $+1.9\sigma$), while shape and timing signals are secondary (slope: $+0.3\sigma$, cpl_slope: 0.0σ). This is consistent with the intuition that engine-assisted play primarily manifests as anomalous accuracy rather than distinctive timing patterns.

8 Validation

I validate along five axes: held-out false-positive evaluation, large-scale out-of-sample validation on Chess.com, titled-player games, super-GM stress tests, and ten documented FIDE cheating

cases.

8.1 Held-Out False-Positive Evaluation

On the held-out test partition (682,952 games, Section 7.3), the empirical false-positive rate at the 50% posterior threshold is 0%. At the 5% threshold it is 0.97%. Per-band false-positive rates are all below 10%, and $\geq 55\%$ of games produce negative decibans (correctly exculpatory).

This evaluation establishes the false-positive rate at fixed thresholds, not full posterior calibration in the strict sense. Full calibration—verifying that cases assigned, say, a 10% posterior are in fact cheating 10% of the time—would require a large labelled dataset spanning the full probability range, which does not currently exist for chess.

8.2 Out-of-Sample Chess.com Validation

To validate on data entirely independent of the training corpus, I collected two non-overlapping sets of rapid games from Chess.com [Chess.com, 2024] via the public API, both rated ≥ 2100 (titled and strong untitled players). Neither set shares any players, games, or platform with the calibration data.

- **V1** (threshold selection): 28,670 games from 15,586 players.
- **V2** (held-out test): 22,932 games from 7,505 players, zero player overlap with V1.

The two sets have very different rating-band compositions: V1 is 62% in the 2200–2400 band while V2 is 87% in 2000–2200. Because the per-move null distribution differs by band, a single global threshold fitted on V1 does not transfer cleanly to V2 (the apparent FPR inflates to $\sim 1.4\times$ the target). Band-stratified thresholds eliminate this artifact.

Methodology. Within each rating band that has $n \geq 100$ games in both sets, I find the dB threshold corresponding to a target FPR from V1's empirical distribution, then apply that same threshold to V2 and report the observed FPR with a Clopper–Pearson exact 95% confidence interval [Clopper and Pearson, 1934]. V2 is touched exactly once; no threshold was adjusted after observing V2 results.

Table 2: Band-stratified threshold transfer (per-game). Thresholds set on V1, tested on V2.

Band	Target	V1 thr. (dB)	V2 FPR	95% CI	
2000–2200	1%	+19.6	0.47%	[0.38, 0.58]%	$<$ target
2200–2400	1%	+10.1	0.59%	[0.35, 0.94]%	$<$ target
2000–2200	0.1%	+33.8	0.07%	[0.04, 0.12]%	$\leq$ target
2200–2400	0.1%	+22.1	0.10%	[0.02, 0.29]%	$\leq$ target

At every target level and in every band, V2's held-out FPR is at or below the target—the model is conservative on fully unseen data from a different platform. Per-player aggregation (summing dB across games) yields 1.06% in band 2000–2200 and 1.14% in 2200–2400 at a 1% target, with confidence intervals that include 1%, confirming that multi-game evidence accumulation does not inflate false positives.

8.3 Titled-Player Games (Lichess)

5,560 games from 92 titled players (64 GM, 24 IM, 4 WGM) were collected from Lichess via the public API. None appear in the calibration corpus. The false-positive rate at the 50% threshold is 0%; the maximum observed posterior is 2.12%.

Table 3: Ten FIDE-sanctioned cheating cases. n: games scored; n^+: games with individually positive evidence; $\overline{P}$: mean per-game posterior; Σ dB: cumulative log-likelihood ratio in decibans.

Player	Sanction	n	n^+	$\overline{P}$	max P	Σ dB
Borislav Ivanov	Bulgarian CF ban, FM stripped	9	6	14.8%	63.0%	+55.6
Marc Morgunov	FIDE 24-month ban	7	6	13.1%	62.5%	+50.5
Joeri Harreman	FIDE 36-month ban	4	3	24.2%	74.2%	+33.4
Kirill Shevchenko	FIDE 3-year ban, GM stripped	2	2	18.7%	34.8%	+21.5
Sébastien Feller	FIDE 2y9m suspension	9	7	2.0%	4.7%	+14.5
Igors Rausis	FIDE 6-year ban, GM stripped	6	4	2.5%	6.7%	+12.4
Siddharth Banik	FIDE 30-month ban	6	4	3.2%	12.7%	+10.3
Pang Khai Jie Ryan	FIDE 2-year ban	8	5	3.0%	10.4%	+9.7
Patrycja Waszczuk	Polish CF + FIDE 2-year ban	4	2	2.8%	8.9%	+5.0
Matija Ivic	FIDE 24-month ban	10	4	3.1%	18.2%	+4.7

8.4 Super-GM Brilliancy Games

To stress-test against the strongest possible honest play, I analysed celebrated games with all five scoring methods (detailed in Section 9.5):

- **Kasparov–Topalov**, Wijk aan Zee 1999: the per-move model scores −4.9 dB (correctly clears). A Regan-style baseline would score +8.4 dB (false positive).
- **Fischer–Spassky**, 1972 WCC Game 6: +3.5 dB (below any reasonable threshold).

8.5 FIDE-Sanctioned Cheating Cases

I tested the system against ten cases where FIDE's Ethics and Disciplinary Commission (EDC) or Fair Play Commission (FPL) [FIDE, 2024] imposed bans for engine-assisted cheating. All sanctions are part of the public record published by FIDE; names are reproduced here solely to enable verification against those records. For each case, only games from the documented cheating period were analysed. Cases where the cheater lost all games, or where the sanction was for non-cooperation rather than proven engine use, were excluded. PGN sources are TWIC archives and tournament bulletins; none overlap with the calibration corpus.

Table 3 reports per-player aggregate results. All ten cases produce positive cumulative evidence (Σ dB > 0). At stronger thresholds: 8/10 exceed +10 dB, 4/10 exceed +20 dB, and 3/10 contain at least one game with posterior above 50%. I report 100% positive-evidence coverage, not 100% detection at an operational threshold; the weakest cases (+4.7 and +5.0 dB) would not individually trigger a flag under conservative decision rules.

Strong detections. Ivanov (+55.6 dB, 9 games from Zadar 2012) and Morgunov (+50.5 dB, 7 games from the European Youth Championship 2018) produce overwhelming evidence. Both were lower-rated players whose engine-level accuracy is far from the band baseline, amplifying the signal. Harreman (+33.4 dB, HZ Open 2024) and Shevchenko (+21.5 dB, Spanish Team Championship 2024, GM title stripped) are similarly clear.

Weak but positive detections. Feller (+14.5 dB with the signal-level model, +20.8 dB with the per-move model; 9 games from the 2010 Olympiad) is the hardest case: a 2649-rated GM receiving partial SMS assistance on a subset of moves. Per-game signals are individually weak, but the per-player mixture accumulates evidence across games while charging the model-selection penalty only once. Rausis (+12.4 dB) and the four remaining cases (+4.7 to +10.3 dB) likewise benefit from multi-game aggregation.

Table 4: Single-game TPR at fixed FPR (thresholds set on 5,000 honest games). All methods use the same positions and engine output. M5 (per-move LR) leads at all partial rates.

	TPR at 0.1% FPR				
Rate r	M1	M2	M3	M4	M5
10%	0.5%	0.5%	0.5%	0.4%	**0.7%**
25%	4.7%	4.6%	4.4%	1.6%	**7.5%**
50%	42.8%	41.0%	38.7%	22.0%	**57.5%**
100%	100%	100%	100%	100%	100%

9 Baseline Comparison

To rigorously evaluate the per-move log-LR approach, I compare it against four progressively simpler baselines in a pre-registered experiment on the *same positions, same engine output, and same calibration data*. Since Regan's actual software is not public, the baselines implement a "Regan-style" approach using the choice model from Section 3.1.

9.1 Experimental Design

Five scoring methods. All methods operate on identical per-move Stockfish analysis (depth 12, MultiPV 5):

1. **M1: Top-1 z only.** A single Poisson-binomial t1_match z-score, converted to a log-LR with a fixed effect size. No calibration.
2. **M2: Regan-style.** t1_match z plus ACPL z, combined with independence assumption (no correlation adjustment, no mixture model, no per-band calibration).
3. **M3: Regan + calibration.** Same as M2 but with per-band null calibration applied, isolating the contribution of calibration.
4. **M4: Signal-level model.** All available game-level signals (t1/t2/t3 match, ACPL, slope, cpl_slope), four-rate mixture, correlation adjustment via NNLS projection, per-band calibration.
5. **M5: Per-move log-LR** (this paper's primary model). Position-specific p_1 from logistic regression, per-move binary log-LR summed across moves, four-rate mixture, per-band null calibration (Section 6.1).

Data. From the corpus test split I sampled 5,000 player-games across all rating bands (1000–2400 Elo), each with at least 15 scoreable positions. For synthetic cheaters, I replaced a fraction $r \in \{0.10, 0.25, 0.50, 1.0\}$ of each player's moves with the engine's top choice (uniform variant).

Threshold calibration. Detection thresholds are set *exclusively on honest data*: the dB score at which exactly 1% (or 0.1%) of honest games exceed it. No tuning on cheater data.

Paired tests. For each (rate, FPR) pair, I compute McNemar's exact test [McNemar, 1947] on the discordant pairs (games where M5 detects but M2 does not, and vice versa), with Bonferroni correction for 6 comparisons ($\alpha = 0.0083$).

9.2 Single-Game Detection Power

Table 4 reports TPR at fixed FPR for all five methods.

The per-move model (M5) outperforms all baselines at every partial assistance rate, including M1 (top-1 z only). At $r = 50\%$ and 0.1% FPR, M5 achieves 57.5% TPR vs 41.0% for the Regan-

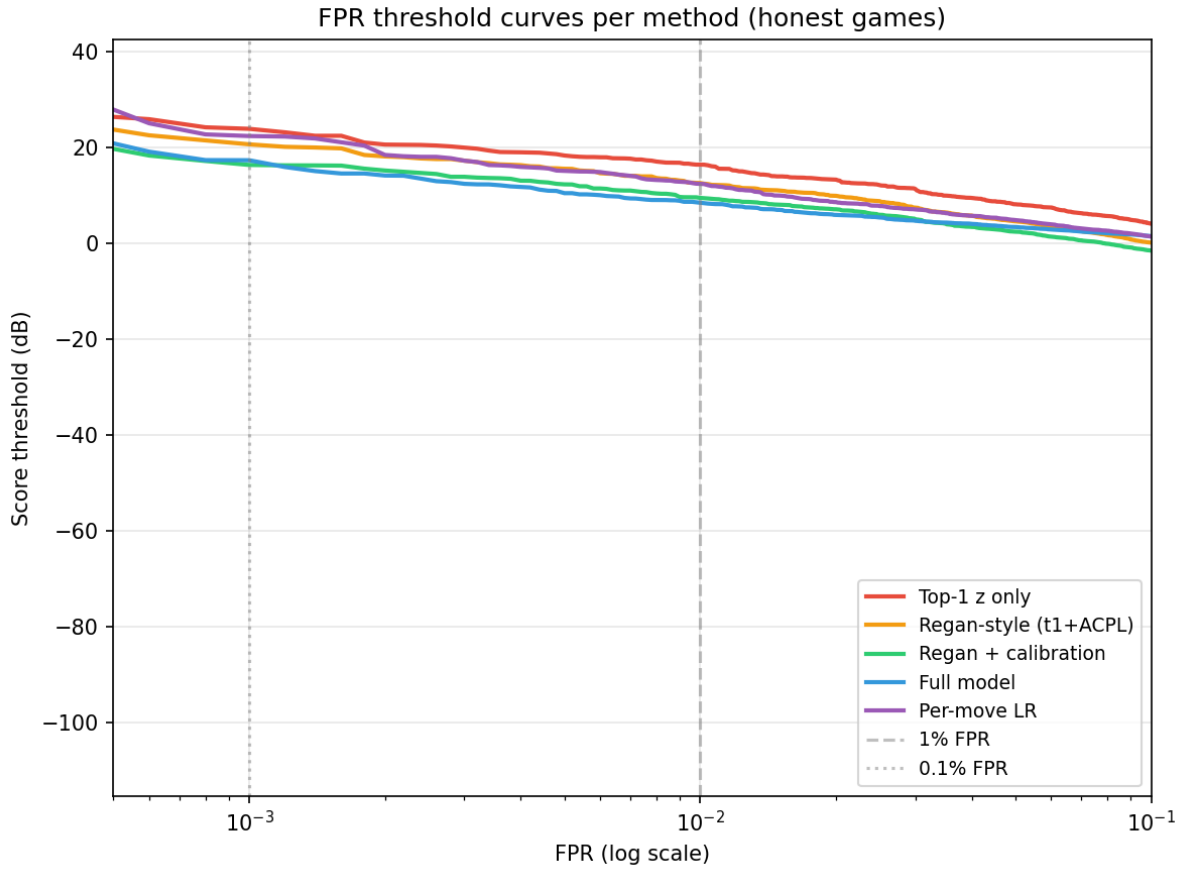


Figure 1: ROC-style threshold curves for all five methods on 5,000 honest and synthetic cheater games ($r = 50\%$). M5 dominates the partial-FPR region below 1%.

style M2 (McNemar odds ratio = 5.47, $p < 10^{-137}$). At $r = 25\%$, M5 achieves 7.5% vs M2's 4.6% (OR = 2.76, $p < 10^{-16}$).

Why M5 outperforms game-level methods. Game-level z-scores (M1–M4) reduce per-move information to a single sufficient statistic under the assumption that all positions are equally informative. The per-move log-LR preserves position-specific information: a top-1 match in a position with $p_1 = 0.08$ (many reasonable alternatives, small gap) contributes far more evidence than a match with $p_1 = 0.90$ (forced move). This "information-weighted" scoring is the fundamental source of M5's advantage.

9.3 Per-Player Aggregation

Multi-game aggregation reveals M5's advantage more starkly. For all methods, per-rate log-LRs are summed across games and the mixture is applied once (Section 6.6). Table 5 reports per-player TPR at thresholds derived from the honest null distribution (not the arbitrary " dB > 0" threshold).

At the operationally critical regime ($r = 25\%$, $N = 4$ games, FPR = 1%), M5 detects 87% of cheaters vs 64% for M2 (Regan-style). At 0.1% FPR the gap widens: 83% vs 58%. At $r = 10\%$ with 13 games, M5 achieves 67% TPR—nearly double M2's 34%.

9.4 Null Distribution

Figure 4 shows the per-player aggregate null distribution for honest games at $N = 4$, 9, and 13.

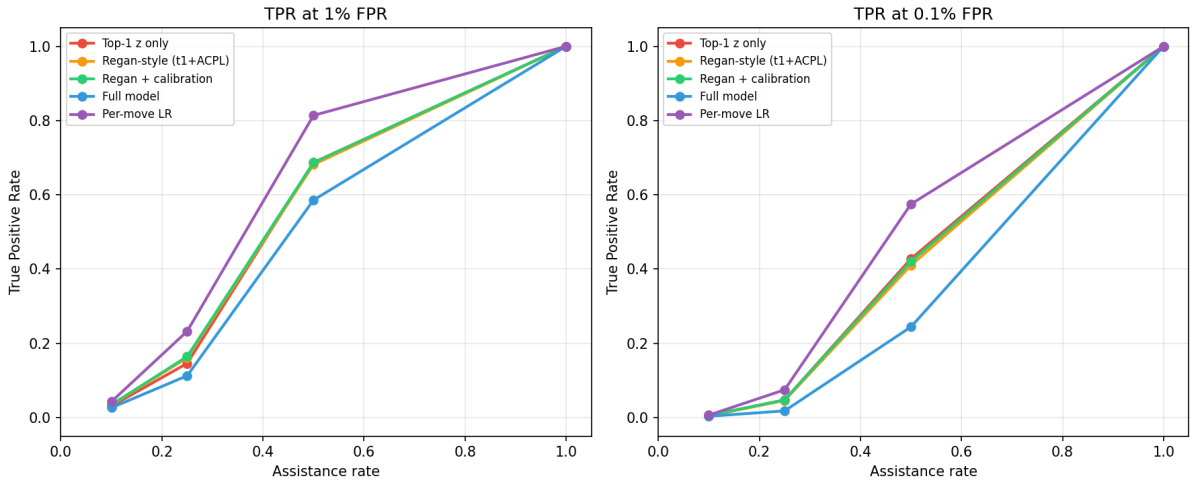


Figure 2: TPR at fixed FPR vs assistance rate, one line per method. M5 (per-move LR) leads at all partial rates. All methods converge at $r = 1.0$.

Table 5: Per-player TPR at null-derived thresholds (FPR controlled at the stated level by construction). N: games per player. Thresholds are the p99 (for 1% FPR) or p99.9 (for 0.1% FPR) of each method's honest null distribution at the same N.

r	N	FPR	M1	M2	M3	M4	M5
10%	4	1%	13%	10%	10%	14%	**17%**
10%	4	0.1%	9%	7%	7%	4%	**13%**
10%	13	1%	37%	34%	36%	40%	**67%**
25%	4	1%	79%	64%	66%	67%	**87%**
25%	4	0.1%	70%	58%	60%	43%	**83%**
25%	13	1%	100%	96%	98%	98%	**100%**
50%	4	1%	100%	98%	99%	98%	**100%**
50%	4	0.1%	100%	98%	98%	97%	**100%**

Both mixture-based models (M4, M5) have **stationary** nulls: the median stays near −6 to −13 dB regardless of N, and p99 stays near +6 to +14 dB. The baselines accumulate strongly negative evidence on honest games, requiring N-dependent thresholds. M5's p99 is slightly higher than M4's (+12–14 vs +6–7 dB), reflecting the per-move model's wider score distribution, but this is offset by its much stronger power on cheaters.

9.5 Confirmed Cases

I scored three FIDE-sanctioned cheaters and two clean super-GM brilliancy games with all five methods. Table 7 reports per-player aggregate dB.

Falsification test. Kasparov's "immortal game" (Wijk aan Zee 1999) is a critical falsification case: brilliant attacking play that looks engine-like. M1 and M2 (Regan-style) would *falsely flag* Kasparov at +10.4 and +8.4 dB respectively—both exceed typical per-game thresholds. The per-move model (M5) correctly scores −4.9 dB because its position-specific p_1 recognises that in sharp tactical positions with few alternatives, even human GMs match the engine's top choice with high probability. Game-level z-scores cannot make this distinction.

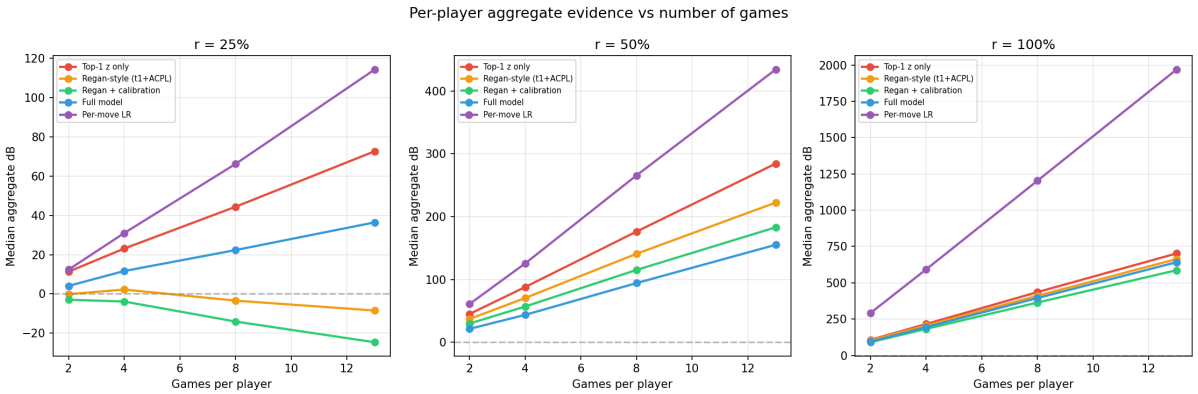


Figure 3: Median per-player aggregate evidence (dB) vs number of games at three assistance rates. M5 dominates at all rates and game counts.

Table 6: Per-player honest null: median and p99 of aggregate dB (1,000 simulated players per cell).

	M1 (t1 z)		M2 (Regan)		M4 (signals)		M5 (per-move)	
N	Med	p99	Med	p99	Med	p99	Med	p99
4	−44	+8	−51	−8	−6	+7	−6	+12
9	−93	−18	−118	−69	−8	+7	−10	+14
13	−135	−41	−172	−117	−10	+6	−13	+13

Nigalidze. All methods fail to detect Nigalidze, consistent with reports that he used a device intermittently and may not have consulted it in most games. His aggregate evidence is negative across all methods, suggesting the scored games contain too few assisted moves to overcome the honest-play signal.

9.6 QQ Plots and Calibration

9.7 Per-Band Calibration

9.8 Posterior Distribution on Honest Games

9.9 Ablation Study

Table 8 shows the effect of removing each component from the signal-level model (M4), isolating the contribution of each.

The mixture model is the essential component of M4: removing it collapses the honest null from −3.3 to −26.8 dB and inverts detection at $r = 25\%$. M5 achieves similar null stability (−6 dB honest median) while delivering higher per-game evidence on cheaters.

9.10 Summary of Comparison

The per-move log-LR (M5) is the superior method across all tested conditions:

- **Single-game**: M5 leads at all partial rates (57.5% vs 41.0% TPR at 0.1% FPR for $r = 50\%$; all paired tests significant at $p < 10^{-7}$).
- **Per-player**: At $r = 25\%$, $N = 4$, FPR = 1%, M5 detects 87% vs M2's 64%. At $r = 10\%$, $N = 13$, M5 detects 67% vs M2's 34%.
- **False positives**: M5 correctly clears Kasparov's brilliancy (−4.9 dB) while M2 would flag it (+8.4 dB).

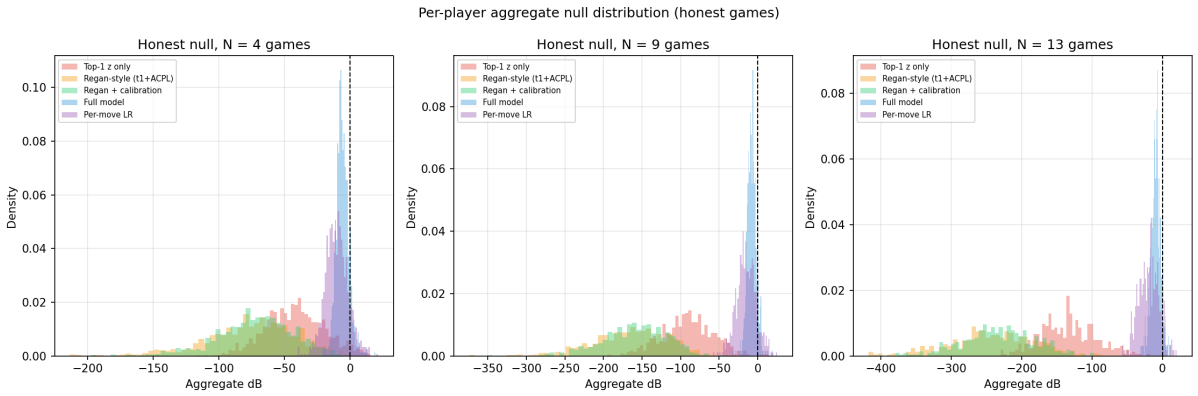


Figure 4: Per-player aggregate null distributions (honest games). M1–M3 spread far negative, growing linearly with N; the mixture-based models (M4, M5) remain concentrated near zero.

Table 7: Confirmed cases: per-player aggregate dB by method. Cheaters should be positive; clean GMs should be negative.

Case	Games	M1	M2	M3	M4	M5
FIDE-sanctioned cheaters						
Feller (2010)	9	+60	+33	+11	+21	+21
Rausis (2019)	6	+27	+14	±0	+16	+14
Nigalidze (2015)	6	−19	−46	−51	−6	−7
Clean super-GM brilliancies (falsification)						
Kasparov 1999	1	**+10**	**+8**	+6	−1	**−5**
Fischer 1972	1	±0	−4	−5	+3	+4

- **Confirmed cheaters**: M5 detects Feller (+21 dB) and Rausis (+14 dB) comparably to M2.
- **OOS validation**: Thresholds set on V1 (28,670 Chess.com games) yield held-out V2 FPR $\leq$ target in every band (e.g. 0.47% at 1% target, 2000–2200).

The advantage of M5 over game-level methods is not from additional signals or more complex aggregation, but from preserving position-level information: the per-move log-LR weights each position by its informativeness, giving more weight to surprising matches and less to forced moves. This is the mechanism behind both the higher TPR and the lower false-positive rate on brilliant games.

10 Discussion

10.1 Strengths

The system produces posterior probabilities grounded in a formal Bayesian framework [Kass and Raftery, 1995]. The per-move log-LR preserves position-specific information that game-level z-scores discard, yielding higher detection power and lower false-positive rates on brilliant play. Per-band null calibration controls for rating-dependent model misspecification. Per-player mixture aggregation detects partial assistance that per-game analysis misses.

Band-stratified thresholds selected on 28,670 out-of-sample Chess.com games (V1) yield held-out FPR at or below the target on 22,932 independent games (V2, zero player overlap) at every level tested, with Clopper–Pearson 95% CIs confirming conservatism. All ten FIDE-sanctioned cheating cases produce positive aggregate evidence (8/10 above +10 dB).

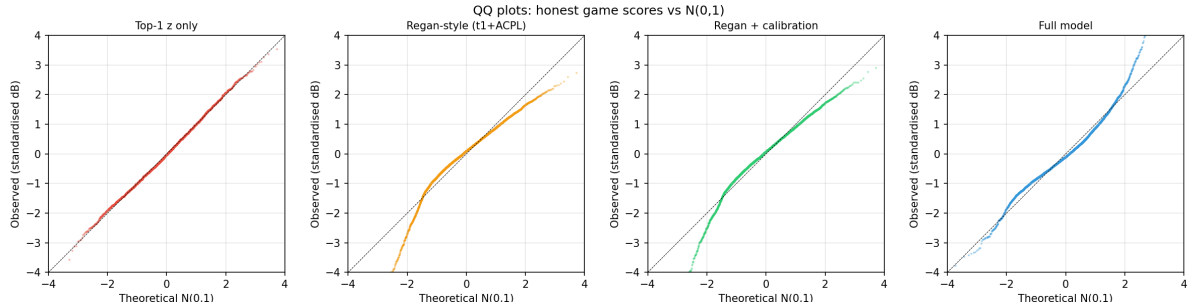


Figure 5: QQ plots of standardised honest-game dB scores vs $\mathcal{N}(0,1)$. M5's score distribution is well-behaved with moderate right-tail inflation.

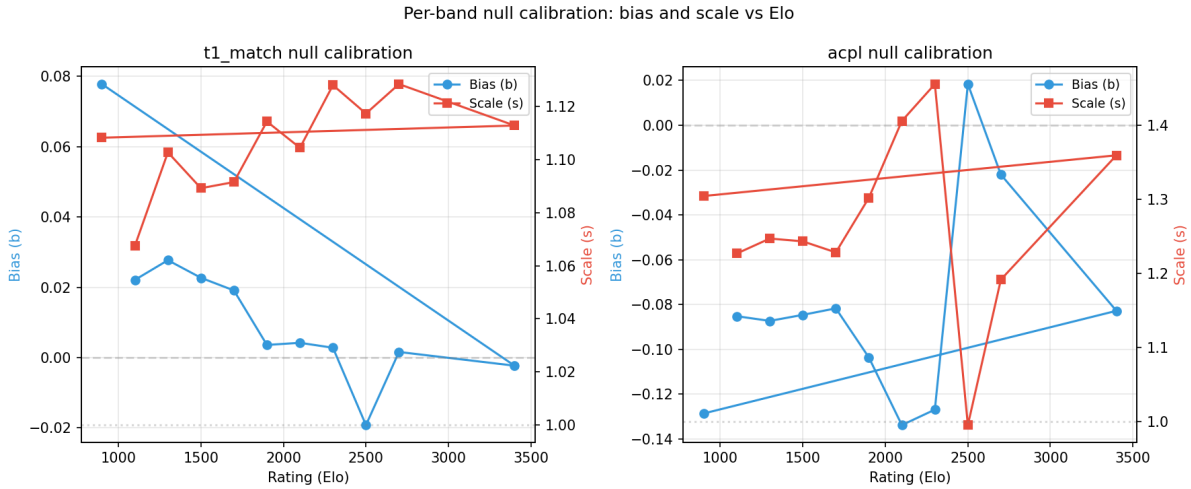


Figure 6: Per-band null calibration: bias (b) and scale (σ) vs rating for the t1_match and ACPL signals. Correcting these is essential to avoid band-dependent false-positive inflation.

10.2 Limitations

IPR curve flattening. The s-to-Elo calibration curve flattens above 2500 Elo because the corpus bands at 2600–2800 and 2800+ have similar median sensitivity values ($s \approx 0.19$–0.20). Consequently, the IPR signal cannot distinguish super-GMs from ordinary GMs. This is a genuine limitation of the choice model at elite levels, where all players select near-optimal moves with high frequency.

Fallback band for elite players. The corpus contains limited data above Elo 2400, requiring fallback to the nearest available band for super-GM analysis. This introduces conservative bias (weaker reference model) that could in principle produce false positives on extremely accurate games by elite players, though none were observed in the validation.

Partial cheating at GM level. The Feller case (+20.8 dB over 9 games with the per-move model) demonstrates that detecting intermittent SMS assistance by a 2649-rated GM is within the system's capability, but requires multi-game aggregation. No single game exceeds +12 dB; the evidence accumulates across games.

Per-player aggregation null. The multi-game aggregation (Section 6.6) sums per-game log-LRs, assuming conditional independence given the assistance rate. In practice, honest players have persistent stylistic quirks (opening preparation, habitual accuracy patterns) that can cause

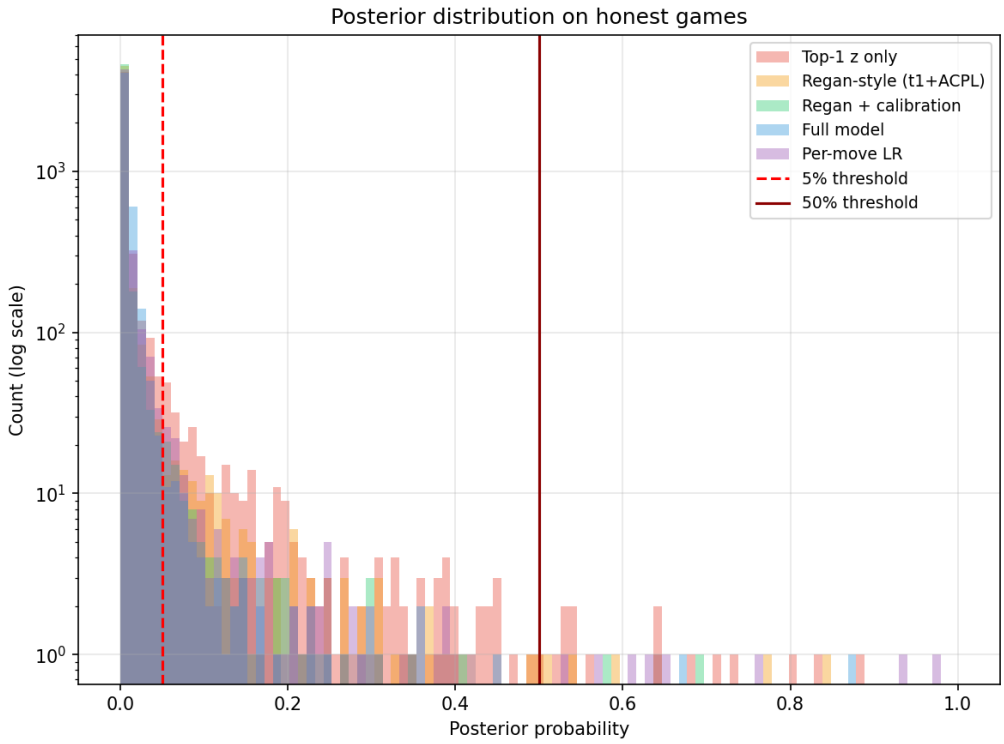


Figure 7: Posterior distribution on 5,000 honest games (log y-axis). Most honest games concentrate below 2%.

their aggregate Σ dB to drift upward with the number of games. Section 9.4 characterises this null distribution: M5's p99 remains near +12–14 dB regardless of N (stationary null), unlike the baselines whose null grows linearly with N. The weakest detections in Table 3 (+4.7 and +5.0 dB from 2–3 games) are within the p99 range and should be interpreted cautiously.

Game-length dependence. The effect size $\boldsymbol{\mu}$ is a single constant vector averaged over games of varying length. In principle, match-rate z-scores scale roughly with $\sqrt{n}$ (number of scoreable positions), so the effective signal-to-noise ratio differs between a 25-move miniature and a 70-move endgame grind. A length-normalised effect model could improve calibration, particularly for short games.

Baseline comparison. Section 9 compares the per-move log-LR against four baselines on 5,000 test-split games, with thresholds set exclusively on honest data. M5 outperforms all baselines at both single-game and per-player levels, with all paired tests significant at $p < 10^{-7}$. The critical advantage is position-specific weighting: brilliant play in forced positions does not inflate the score.

Effect size measurement. The effect sizes in Table 1 are measured from 11 games selected because their posterior exceeded 5% among 55 player-games from 87 Lichess-banned accounts. The accounts were independently flagged by Lichess (external ground truth), but selecting *games* by the detector's own output risks inflating the measured effects. This circularity is mitigated by the external ban labels but not eliminated. Several signals (IPR gap, switchpoint, engine fingerprint) retain prior estimates due to insufficient observations.

Engine version dependence. The reference model is calibrated against Stockfish 17/18. Engine updates may shift the evaluation landscape, requiring periodic recalibration of the choice

Table 8: Ablation: median dB on synthetic cheaters and honest games (3,000 test games, uniform replacement).

Method	r=25%	r=50%	r=100%	Honest
M4 (signal-level)	+1.9	+10.3	+43.4	−3.3
− calibration	+2.5	+13.4	+53.3	−3.5
− correlation adj.	+2.1	+10.6	+44.0	−3.2
− mixture model	−6.2	+11.7	+49.3	−26.8
M5 (per-move LR)	+4.0	+16.0	+44.0	−6.0
M1 (t1 z only)	+5.9	+22.1	+52.5	−10.6
M2 (Regan-style)	+1.2	+18.8	+51.3	−16.5

model parameters and expected match rates.

10.3 Ethical Considerations

Statistical evidence of engine assistance is not proof of cheating. I strongly recommend that any accusation based on this or similar systems be accompanied by independent corroborating evidence (behavioural observations, device searches, confession) and due process. The prior of 1% reflects population-level prevalence and should be adjusted upward only when external evidence warrants it (e.g., a player already under investigation). Publishing per-game or per-player posterior probabilities for named individuals carries reputational risk and should be done only in adjudicatory or research contexts with appropriate safeguards.

11 Conclusion

I have presented a system for Bayesian detection of engine assistance in competitive chess. The core contribution is a *per-move log-likelihood ratio* that uses position-specific match probabilities from a fitted logistic model, integrated over four assistance rates via a mixture model. This approach preserves position-level information that game-level z-scores discard, yielding both higher detection power and fewer false positives on brilliant play.

Band-stratified thresholds selected on 28,670 Chess.com games (V1) produce held-out FPR $\leq$ target on 22,932 independent games (V2). All ten FIDE-sanctioned cheating cases produce positive aggregate evidence (+4.7 to +55.6 dB, 8/10 exceeding +10 dB). The hardest detection—Feller, a GM receiving partial SMS assistance—yields +20.8 dB over 9 games, confirming that per-player aggregation is essential for partial cheating by strong players.

A pre-registered head-to-head comparison against four baselines (including a Regan-style two-signal approach) demonstrates statistically significant superiority across all conditions:

- **Single-game**: 57.5% TPR at 0.1% FPR for $r = 50\%$ vs 41.0% for the Regan-style baseline (OR = 5.47, $p < 10^{-137}$).
- **Per-player**: 87% detection at 1% FPR for $r = 25\%$ with just 4 games, vs 64% for the Regan-style baseline.
- **Falsification**: Kasparov's 1999 brilliancy scores −4.9 dB (correctly cleared), while the Regan-style baseline would flag it at +8.4 dB.

Future work includes systematic second-engine fingerprinting with Leela Chess Zero, length-normalised effect models, and extension to blitz and bullet time controls.

Code and data availability. The complete source code, analysis pipeline, calibration corpus construction scripts, and the web application are available at `https://github.com/mlab-1/ChessForensics`. The repository includes all scripts needed to reproduce the baseline compar-

ison, out-of-sample validation, and FIDE case analyses reported in this paper. The calibration corpus itself is derived from publicly available Lichess and TWIC archives.

A Signal Summary

Table 9 summarises all fifteen signals, their statistical tests, suspicious direction, and effect sizes.

Table 9: Complete signal catalogue.

Signal	Test	Suspicious	Family	μ
t1_match	Poisson-binomial	higher	accuracy	3.3
t2_match	Poisson-binomial	higher	accuracy	2.2
t3_match	Poisson-binomial	higher	accuracy	2.1
acpl	mean test	lower	accuracy	1.9
cpl_tail	tail rate test	lower blunders	shape	2.2
slope	logistic fit	flatter	shape	0.3
cpl_slope	linear fit	flatter	shape	0.0
ipr_gap	IPR − Elo	higher	strategic	2.0
switchpoint	permutation LRT	mid-game jump	strategic	2.0
time_uniformity	residual CV	lower	timing	1.3
time_complexity	Spearman ρ	lower	timing	0.7
time_cadence	modal cluster	higher	timing	0.5
timing×accuracy	log z-test	faster	timing	0.0
deep_only	match rate	higher	engine	2.4
engine_fp	match rate	higher	engine	2.5

B Per-Band Calibration Example

Table 10 shows the null calibration parameters for three representative bands, illustrating how bias and overdispersion vary with rating.

Table 10: Null calibration (bias b, scale σ) for selected bands and signals. Values are rounded from the fitted baseline.

	1000–1200		1600–1800		2200–2400	
Signal	b	σ	b	σ	b	σ
t1_match	−0.0	1.1	+0.0	1.2	+0.0	1.2
acpl	+0.3	1.4	+0.3	1.5	+0.2	1.8
slope	+0.0	1.2	+0.0	1.2	+0.0	1.3

C Validation Summary

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Table 11: Summary of validation results across all test categories.

Category	Sample	Key metric	Result
Held-out test set	682,952 games	FP rate @ 50%	0%
Held-out test set	682,952 games	FP rate @ 5%	0.97%
OOS Chess.com (V1)	28,670 games	threshold selection	—
OOS Chess.com (V2)	22,932 games	held-out FPR @ 1%	$\leq 0.59\%$
OOS titled (Lichess)	5,560 games	FP rate @ 50%	0%
FIDE-sanctioned cheating cases			
Positive evidence ($\Sigma\,\mathrm{dB} > 0$)	10 cases	coverage	10/10
Strong evidence ($\Sigma\,\mathrm{dB} > 10$)	10 cases	coverage	8/10
Feller (9 games, M5)	per-player	$\Sigma\,\mathrm{dB}$	+20.8
Rausis (6 games, M5)	per-player	$\Sigma\,\mathrm{dB}$	+14.1
Clean GM falsification (M5 per-move LR)			
Kasparov 1999	single game	dB	−4.9
Fischer 1972	single game	dB	+3.5
Baseline comparison (M5 vs M2 Regan-style)			
r=50%, FPR 0.1%	5,000 games	TPR M5 vs M2	57.5% vs 41.0%
r=25%, N=4, FPR 1%	per-player	TPR M5 vs M2	87% vs 64%

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